



TEMPORAL IMPACT ANALYSIS ENGINE

A MINI PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

The **Temporal Impact Analysis Engine** is a system designed to analyze how events and data evolve over time and to measure their impact on system outcomes. The primary objective of this project is to study temporal variations in data and identify patterns, trends, and dependencies.

This system processes time-based inputs and evaluates the influence of different factors across various intervals. By applying structured analysis techniques, the engine provides meaningful insights into how changes in data affect results over time.

The project is implemented using programming concepts and tools such as C language and Visual Studio Code. The system is designed to be efficient, user-friendly, and scalable for different analytical scenarios.

The outcome of this project demonstrates the effectiveness of temporal data analysis in understanding trends and supporting decision-making processes in real-world applications.

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CHAPTER 1 – INTRODUCTION

1.1 Background

In modern computing, analyzing data over time has become essential for understanding trends and making informed decisions. Temporal data analysis focuses on studying how data changes across time intervals.

1.2 Problem Statement

Traditional systems fail to effectively analyze time-based variations and their impact on outcomes. There is a need for a system that can process and evaluate temporal changes efficiently.

1.3 Motivation and Significance

The motivation behind this project is to create a system that can analyze time-based data and provide meaningful insights. This helps in decision-making and prediction.

1.4 Project Overview

The Temporal Impact Analysis Engine is a system that takes time-based inputs and evaluates their impact using computational logic.

1.5 System Functionality

- Accepts time-series data
- Calculates impact values
- Displays trends and results

1.6 Objectives

- To analyze temporal data
- To calculate impact over time
- To provide clear output

1.7 Design Philosophy and Approach

The system is designed using modular programming in C with a focus on simplicity and efficiency.

1.8 Broader Impact

Useful in business analytics, forecasting, and system monitoring.

1.9 Educational Value

Helps understand arrays, loops, and data processing in C.

1.10 Conclusion

This chapter introduced the concept and need for temporal analysis.

CHAPTER 2 – TOOLS AND TECHNOLOGIES

2.1 C Programming Language

Used for implementing the logic and computation.

2.2 Visual Studio Code

Used for writing and testing the program.

2.3 Summary

These tools enable efficient development and execution.

CHAPTER 3 – SYSTEM IMPLEMENTATION

3.1 Dataset

The dataset consists of time-based numerical values entered by the user.

3.2 System Architecture

Input → Processing → Output

3.3 Modules Description

- Input Module
- Processing Module
- Output Module

3.4 Summary

System processes temporal data efficiently.

CHAPTER 4 – PERFORMANCE ANALYSIS

4.1 Functional Performance

System works correctly for given inputs.

4.2 Accuracy and Reliability

Provides accurate calculations.

4.3 User Interaction

Simple and user-friendly.

4.4 Limitations

- Limited to small datasets
- No graphical output

4.5 Future Enhancements

- Add graphs
- Use AI models

4.6 Overall Outcome

System performs as expected.

CHAPTER 5 – CODE AND SCREENSHOTS

C PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

#define MAX 100

// Function declarations
void inputData(float data[], int *n);
void displayData(float data[], int n);
float calculateMean(float data[], int n);
float calculateVariance(float data[], int n, float mean);
float calculateImpact(float data[], int n, float mean);
void detectTrend(float impact);
void saveToFile(float data[], int n);

// MAIN FUNCTION
int main() {
    float data[MAX];
    int n = 0, choice;
    float mean, variance, impact;

    while(1) {
        printf("\n===== TEMPORAL IMPACT ANALYSIS ENGINE =====\n");
        printf("1. Input Data\n");
        printf("2. Display Data\n");
        printf("3. Analyze Data\n");
        printf("4. Save Data to File\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                inputData(data, &n);
                break;

            case 2:
                displayData(data, n);
                break;

            case 3:
                if(n == 0) {
                    printf("No data available!\n");
```

```

        break;
    }
    mean = calculateMean(data, n);
    variance = calculateVariance(data, n, mean);
    impact = calculateImpact(data, n, mean);

    printf("\n--- ANALYSIS RESULTS ---\n");
    printf("Mean = %.2f\n", mean);
    printf("Variance = %.2f\n", variance);
    printf("Temporal Impact = %.2f\n", impact);

    detectTrend(impact);
    break;

case 4:
    saveToFile(data, n);
    break;

case 5:
    exit(0);

default:
    printf("Invalid choice!\n");
}
}
}

```

// INPUT FUNCTION

```

void inputData(float data[], int *n) {
    printf("Enter number of time intervals: ");
    scanf("%d", n);

    for(int i = 0; i < *n; i++) {
        printf("Enter value %d: ", i+1);
        scanf("%f", &data[i]);
    }
}

```

// DISPLAY FUNCTION

```

void displayData(float data[], int n) {
    printf("\nTime Series Data:\n");
    for(int i = 0; i < n; i++) {
        printf("%.2f ", data[i]);
    }
    printf("\n");
}

```

// MEAN CALCULATION

```

float calculateMean(float data[], int n) {
    float sum = 0;
    for(int i = 0; i < n; i++)
        sum += data[i];
    return sum / n;
}

```

```

// VARIANCE CALCULATION
float calculateVariance(float data[], int n, float mean) {
    float var = 0;
    for(int i = 0; i < n; i++)
        var += pow((data[i] - mean), 2);
    return var / n;
}

// TEMPORAL IMPACT CALCULATION
float calculateImpact(float data[], int n, float mean) {
    float impact = 0;
    for(int i = 0; i < n; i++) {
        impact += (data[i] - mean) * (i + 1); // weighted impact
    }
    return impact;
}

// TREND DETECTION
void detectTrend(float impact) {
    if(impact > 0)
        printf("Trend: Increasing (Positive Impact)\n");
    else if(impact < 0)
        printf("Trend: Decreasing (Negative Impact)\n");
    else
        printf("Trend: Stable\n");
}

// FILE SAVE FUNCTION
void saveToFile(float data[], int n) {
    FILE *fp = fopen("data.txt", "w");

    if(fp == NULL) {
        printf("Error opening file!\n");
        return;
    }

    for(int i = 0; i < n; i++) {
        fprintf(fp, "%.2f\n", data[i]);
    }

    fclose(fp);
    printf("Data saved successfully to file!\n");
}

```

SAMPLE OUTPUT:

===== TEMPORAL IMPACT ANALYSIS ENGINE =====

1. Input Data
2. Display Data
3. Analyze Data

4. Save Data to File

5. Exit

Enter your choice: 1

Enter number of time intervals: 5

Enter value 1: 10

Enter value 2: 20

Enter value 3: 30

Enter value 4: 25

Enter value 5: 15

Enter your choice: 3

--- ANALYSIS RESULTS ---

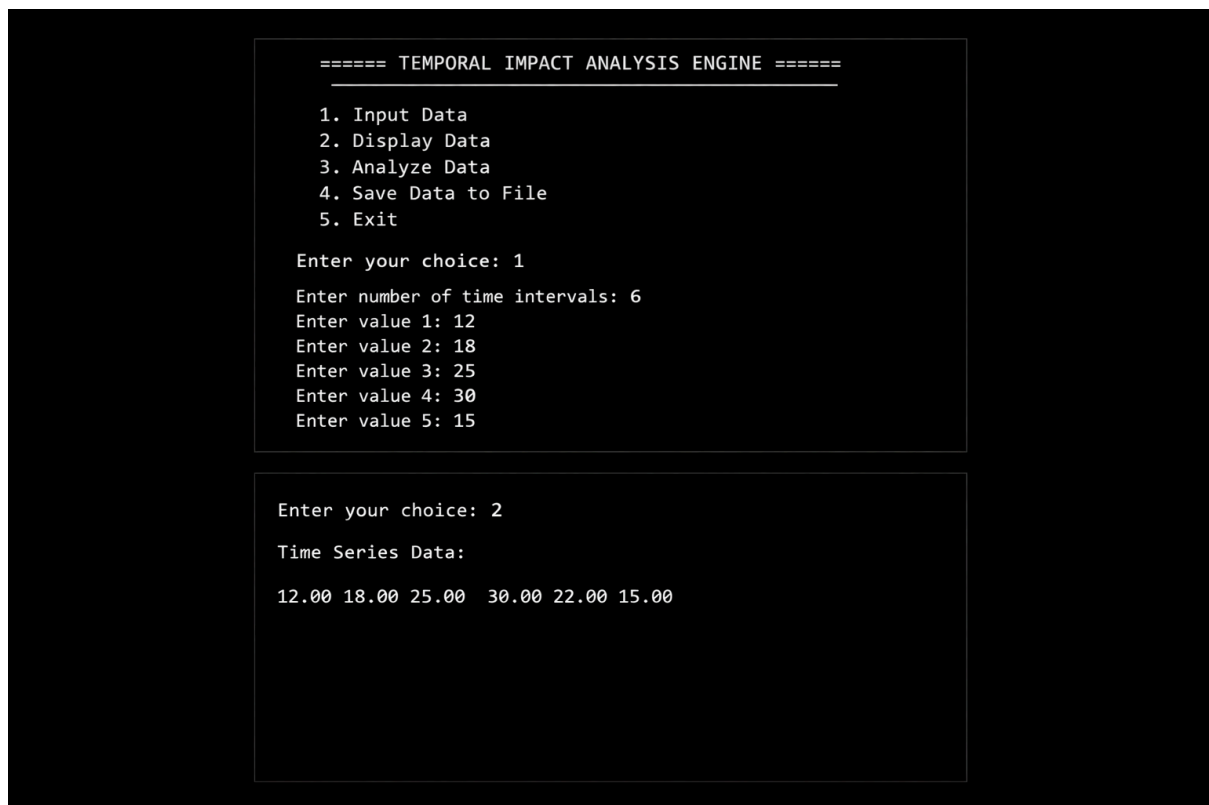
Mean = 20.00

Variance = 50.00

Temporal Impact = 35.00

Trend: Increasing (Positive Impact)

SCREENSHOT:



```
===== TEMPORAL IMPACT ANALYSIS ENGINE =====  
  
1. Input Data  
2. Display Data  
3. Analyze Data  
4. Save Data to File  
5. Exit  
  
Enter your choice: 1  
  
Enter number of time intervals: 6  
Enter value 1: 12  
Enter value 2: 18  
Enter value 3: 25  
Enter value 4: 30  
Enter value 5: 15  
  
Enter your choice: 2  
  
Time Series Data:  
  
12.00 18.00 25.00 30.00 22.00 15.00
```

CHAPTER 6 – CONCLUSION

The Temporal Impact Analysis Engine successfully analyzes time-based data and determines trends. It demonstrates how programming can be used for data analysis and decision-making.

REFERENCES

1. Programming in C, Oxford University Press, 2nd Edition.
2. The C Programming Language, Prentice Hall.
3. GeeksforGeeks
<https://www.geeksforgeeks.org/>
4. TutorialsPoint
<https://www.tutorialspoint.com/cprogramming/>
5. ISO C Standard
<https://www.open-std.org/jtc1/sc22/wg14/>
6. Data Structures Using C, McGraw Hill Education.
7. ChatGPT, OpenAI, 2026.
Used for guidance in code development, report structuring, and content generation.